

SNEHA PANDIT

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RESEARCH INTERESTS

- Human-AI Interaction, Multimodal AI Systems, NLP, Behavioral Modeling, AI Safety & Evaluation

EDUCATION

GLA University *B.Tech in Computer Science & Engineering (Spec. AI & ML)* | **Expected:** 2027

- **Core Coursework:** Advanced Machine Learning, Natural Language Processing, Deep Learning, Data Structures & Algorithms, OS.

RESEARCH & PROJECT EXPERIENCE

Sentia: Multimodal Emotion Drift Analytics Platform

[GitHub](#) | [Technical Report](#)

→ *Python, PyTorch, Hugging Face, RoBERTa, FastAPI, WebSockets*

- **Tracked emotional drift across multi-turn user sessions** using probabilistic distance metrics (TVD and Euclidean shifts) over 7-class emotional distributions to model emotional drift patterns across long-horizon interactions.
- **Designed evaluation workflows** to analyze behavioral consistency, emotional drift, and interaction safety patterns across long-horizon human-AI conversations.
- **Improved multi-turn classification accuracy** over text-only baselines by engineering a confidence-weighted late-fusion architecture combining text embeddings with a facial expression CNN.
- **Fine-tuned a RoBERTa-base transformer** on a normalized 7-class schema of the GoEmotions dataset (58k samples) to achieve a macro F1-score above 0.70.
- **Developed a clause-level text segmentation pipeline** to detect intra-message emotional transitions in conversational inputs.
- **Architected a 7-stage parallel inference pipeline** executing concurrent text and vocal feature extraction loops via a ThreadPoolExecutor to maintain P95 latency below 4 seconds.

Panopticon: Distributed Political-Economic Data Observability Engine

[GitHub](#) | [Technical Report](#)

→ *TypeScript, Node.js, Express, Prisma, BullMQ*

- **Integrated an isolated LLM processing layer** via background workers to parse unstructured news feeds, automate entity extraction, and dynamically update relational data fields.
- **Engineered a distributed tracing backend** using TypeScript and Prisma ORM to aggregate and map multi-node political and economic data models.
- **Developed asynchronous data-ingestion workers** via BullMQ to process 10k+ structured events/day across automated cron cycles and task queues.
- **Designed event-driven diagnostic telemetry loops** using WebSockets to broadcast low-latency system-health metrics to a central monitoring cockpit.

TECHNICAL SKILLS

- **AI/ML & Research:** Multimodal AI Systems, Transformer Models, NLP, Human-AI Interaction Modeling, Deep Learning, PyTorch, Hugging Face Transformers, RoBERTa, Behavioral Signal Analysis.
- **Languages:** Python, TypeScript, JavaScript, SQL, Java, C++.
- **Systems & Infrastructure:** FastAPI, Node.js, Express, AsyncIO, Redis, Prisma ORM, RESTful APIs, WebSockets, MongoDB, JWT Authentication, Docker, GitHub Actions.

OPEN-SOURCE CONTRIBUTIONS & LEADERSHIP

Project Admin / Contributor | GirlScript Summer of Code (GSSoC) | 2025 – 2026

- **Project Admin (2026):** Moderating repository code health, structuring detailed contribution guidelines, and reviewing/merging Pull Requests across deep learning and backend optimization codebases.
- **Contributor (2025-2026):** Implemented modular feature extensions, resolved Git branch conflicts, and refined processing scripts across open-source AI/ML repositories.

AWARDS & ACHIEVEMENTS

- **National Finalist** | Smart India Hackathon (SIH), Ministry of Education, Govt. of India | **2024**
 - *Project Theme: Smart Land Governance AI Engine.* Selected to compete at the 36-hour Grand Finale; engineered the core asynchronous backend framework and model validation integration layer.
- **University Selection Lead** | Smart India Hackathon Internal Nominee | **2024**
 - Secured 1st Place in campus-wide preliminaries out of all competing teams at GLA University.
- **National Phase 2 Selection** | IBM Praambh Hack Challenge | **2025**
 - Advanced through competitive national screening evaluating technical system architecture and machine learning pipeline scalability.

